

CryptoLens

IPsec VPN Security Assessment Report

Parameter	Value
Job ID	job_d5b2ebe5
Assessment Status	COMPLETED
Risk Score	75.00 / 100
Risk Level	MODERATE
AI Confidence	94.00%
Heuristic / AI Agreement	Yes

1. Executive Summary

CryptoLens assessed the supplied IPsec traffic with an overall security score of 75.00/100, classified as MODERATE. The assessment combines protocol evidence, configured scoring rules, and the analyzer outputs available for this job.

3 finding(s) were generated during the assessment.

2. Control-Plane Assessment

Parameter	Value
IKE Version	IKEv2
Operating Mode	Tunnel
Encryption Algorithm	AES-128-CBC
Integrity Algorithm	HMAC-SHA256
DH Group	14
PFS	Disabled
Key Lifetime (seconds)	28800
Replay Protection	Enabled

3. Score Breakdown

Category	Score	Maximum	Utilization
Encryption	15.00	25.00	60.0%
Integrity	15.00	15.00	100.0%

Category	Score	Maximum	Utilization
Key Exchange	10.00	15.00	66.7%
PFS	0.00	10.00	0.0%
Replay Protection	10.00	10.00	100.0%
Key Lifetime	10.00	10.00	100.0%
IKE Version	10.00	10.00	100.0%
Operating Mode	5.00	5.00	100.0%

4. Security Findings

Finding ID	Severity	Category	Observed Value
ENC_001	MEDIUM	Encryption	AES-128-CBC
KEX_001	MEDIUM	Key Exchange	14
PFS_DISABLED	HIGH	Pfs	false

1. Suboptimal Encryption Cipher

Description: AES-128-CBC is acceptable but AES-GCM is recommended for modern deployments.

Source: compliance_map.yaml

2. Legacy DH Group

Description: DH Group 14 (2048-bit MODP) is acceptable but ECP groups (19, 20, 21) are preferred.

Source: compliance_map.yaml

3. PFS Disabled

Description: Perfect Forward Secrecy is disabled. Past traffic may be decrypted if long-term keys are compromised.

Source: compliance_map.yaml

5. Traffic Analysis

Heuristic prediction: Tunnel

AI prediction: Tunnel

Traffic Type	Percentage	Packets	Avg. Packet Size (bytes)
VoIP	72.40%	1240	160.00

6. Standards Compliance Assessment

NIST SP 800-77 Rev. 1

Overall Status	Aligned	Review	Fail	Not Assessed
REVIEW	4	3	0	0

Control	Status	Observed Value	Assessment
ike_version	ALIGNED	IKEv2	IKE version used for IPsec Security Association establishment.
encryption	REVIEW	AES-128-CBC	Observed IPsec confidentiality algorithm.
integrity	ALIGNED	HMAC-SHA256	Observed integrity/authenticated-encryption mechanism.
key_exchange	REVIEW	14	Observed IKE Diffie-Hellman group.
pfs	REVIEW	Disabled	Whether Perfect Forward Secrecy is configured for the applicable child SAs.
replay_protection	ALIGNED	Enabled	Whether IPsec replay protection is enabled.
operating_mode	ALIGNED	Tunnel	IPsec operating mode.

Commercial National Security Algorithm Suite 2.0

Overall Status	Aligned	Review	Fail	Not Assessed
REVIEW	2	5	0	0

Control	Status	Observed Value	Assessment
encryption	REVIEW	AES-128-CBC	CNSA 2.0 confidentiality algorithm assessment.
integrity	REVIEW	HMAC-SHA256	CNSA 2.0 integrity/hash assessment where applicable.
key_exchange	REVIEW	14	CNSA/CSfC key-establishment assessment. Applicability is package-specific.
ike_version	ALIGNED	IKEv2	IKE version assessment for the applicable IPsec profile.
pfs	REVIEW	Disabled	Forward-secrecy assessment where required by the applicable profile.
replay_protection	ALIGNED	Enabled	Replay-protection assessment.
operating_mode	REVIEW	Tunnel	Mode suitability is deployment/package dependent.

6. Assessment Metadata

Parameter	Value
Job ID	job_d5b2ebe5
Status	completed
Processed Packets	1240
AI Confidence	94.00%
Heuristic / AI Agreement	Yes